

Over the past 75 years the study of kaon physics has been integral to the development of the standard model. Since 2016, the NA62 experiment has been the leading experiment in the field of kaon physics. However, the future of kaon physics is uncertain, with the NA62 experiment funded up to CERN LS3, which is expected at the end of 2026. It was then required to investigate the future of kaon physics and the potential experiments that could succeed NA62. This chapter will discuss the potential future of kaon physics, the experiments that could be used to further the field, and the work contributed to these experiments.

0.1 High Intensity Kaon Experiments (HIKE) Description

The High-intensity kaon experiments (HIKE) was a proposed project to continue kaon physics at CERN in the ECN3 experimental hall and represented a long-phased plan after LS3. HIKE would have profited from a four to six-fold increase in beam intensity compared to NA62 and included cutting-edge detector technologies that will be described in this section. There were three phases within the HIKE program, the first being a continuation of NA62's charged kaon physics with the high-intensity beam and detector upgrades to accommodate the rate increase. The second would have involved neutral kaon physics with charged secondary particle tracking, and the third phase would have been designed to measure the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay.

0.1.1 Phase 1

The experimental design of NA62 has been proven as a successful setup for the precision measurement of the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay. The HIKE Phase 1 detector setup is shown in Figure 1 with new and upgraded detectors with respect to the NA62 setup described in Section ?? . The new and upgraded detectors were required due to the four times greater particle rate with respect to NA62 equating to 1.2×10^{13} protons per spill on the target. The expected beam had similar characteristics to NA62 with a $75 \text{ GeV}/c$ secondary beam produced from the $400 \text{ GeV}/c$ primary beam from the SPS accelerator onto a target at an angle of 0 rad .

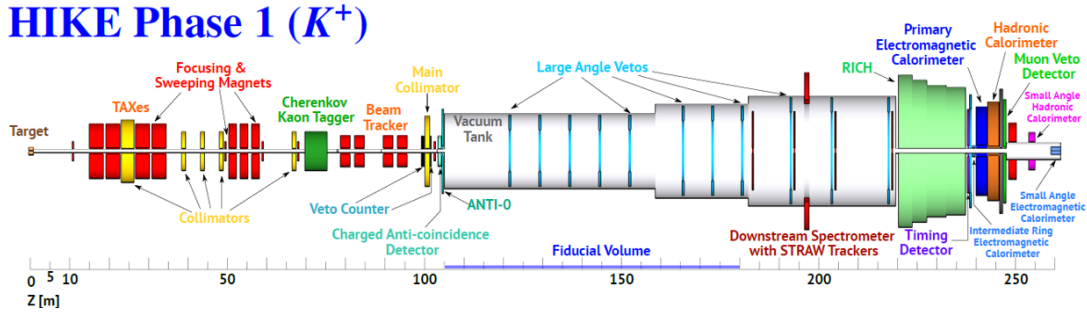


Figure 1: Diagram of HIKE phase 1 detector setup [1]

The upgraded detectors before the fiducial volume included an upgraded photon-detection system for the KTAG, described in section 0.2 and a new GTK chip design described in section 0.3. It was proposed to upgrade the existing NA62 Vetocounter with scintillating fibre (SciFi) technologies to improve the granularity and to accommodate the increased beam intensity.

Major upgrades were envisioned to the LAV and STRAW detectors within the fiducial volume. A number of the LAV stations would have been upgraded to a new design proposed for a lead scintillator design, similar to the Vacuum Veto System (VVS) planned to be used at the CKM experiment [2, 3]. The LAV detectors were

important for the next phases of HIKE, especially for Phase 3 and would have been designed for the requirements of the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ measurement even when used in this Phase. The STRAW spectrometer would have been upgraded to use a new 5 mm straw design shown in Figure 2. The reduction in straw diameter would have decreased drift times and increase granularity resulting in an expected improvement in the track angle and momentum resolution of 10-20 % compared to NA62.

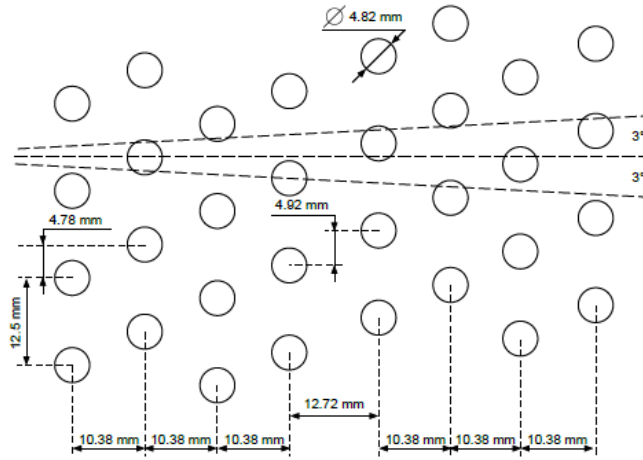


Figure 2: Diagram of the expected layout of the 5 mm straws [1]

Downstream of the fiducial volume, the PMT arrays of the RICH detector would have been upgraded, which is described in Section 0.4. The timing detector would profit from the current NA62-CHOD detector with upgraded PMTs, which would have been chosen from the number of studies from other detectors such as KTAG, RICH and MUV. For Phase 1, the primary electromagnetic calorimeter would have used the LKr from NA62, which had excellent energy resolution but did not satisfy the timing requirements. The required timing resolution would have been achieved by upgrading the readout system. A number of other detectors would have been upgraded or replaced with higher granularity, such as the hadronic calorimeter, MUVs, IRC, and SAC.

It was expected that Phase 1 would run between LS3 and LS4, with the exact years being undetermined. During Phase 1 the expected protons per spill was 1.3×10^{13} (four times NA62) with an nominal year consisting of 200 data taking days with an average of 3000 spills per day. Which with a kaon decay in the fiducial volume per proton on target of 2.5×10^{-6} would result in an expected 2×10^{13} kaon decays per year. At this rate it would be expected that the Phase 1 goals could be completed in 5 years of data taking.

0.1.2 Phase 2

HIKE phase 2 would have been a multi-purpose neutral K_L decay experiment with the primary goal to measure the branching ratios of the $K_L \rightarrow \pi^0 l^+ l^-$ decays. The detector setup for phase 2, shown in Figure 3 would have involved minimal modification to the experimental setup of Phase 1, with the removal of detectors that were only applicable to the charge kaon beam. The KTAG, GTK, RICH, and SAC detectors would have been removed the straw spectrometer shortened to account for the softer momentum of the secondary particles, and the chambers would have been reconfigured to have central beam holes to account for the neutral beam. The MNP33 magnet required a reduction in strength by 20 % to account for the shorter spectrometer, resulting in a momentum kick of $210 MeV/c$. Due to the compact detector, the fiducial volume would have been increased to 90m compared to 75m in Phase 1.

HIKE Phase 2 (K_L)

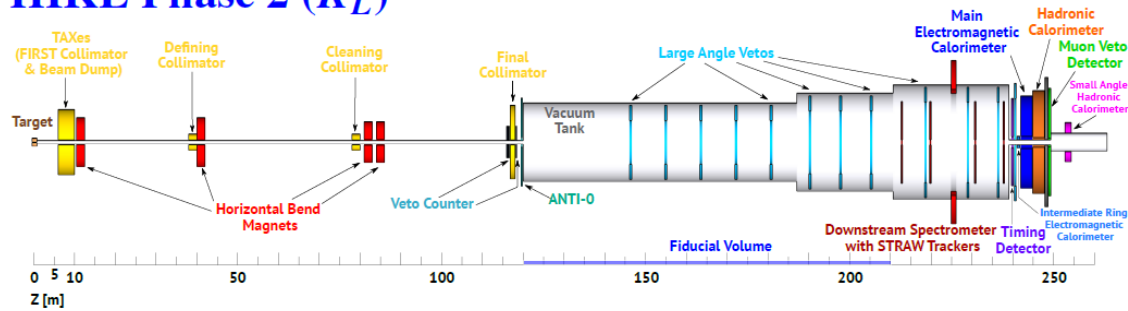


Figure 3: Diagram of HIKE phase 2 detector setup [1]

The design proposed a production angle of 2.4 mrad on the target producing an expected K_L yield of 5.4×10^{-5} per proton. Assuming an integrated proton flux of 1.2×10^{19} protons on target per year would have resulted in 3.8×10^{13} K_L decays in the fiducial volume per year. The mean momentum at the entry of the fiducial volume was expected to be $79 \text{ GeV}/c$ with a mean momentum for K_L which decay of $46 \text{ GeV}/c$.

Phase 2 would be expected to follow after Phase 1 which would of been expected after LS4 up to LS5. The expected protons per spill was six times that of NA62 at 2×10^{13} . With a similar standard year as Phase 1 and an expected 3.2×10^6 K_L per proton on target would result in 3.8×10^{13} decays per year. The expected time to Phase 2 goals was 6 years of standard data taking.

0.1.3 Phase 3 (KLEVER)

HIKE phase 3, previously known as KLEVER [4], was an experiment designed to measure the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay. The experimental setup is shown in Figure 4 and differs from the design used at NA62 and HIKE Phase1/2. The setup consists of a large number of photon-detectors, located around a 160 m long vacuum volume to produce a complete coverage of photons produced from K_L decays. The photon-

detectors include coverage for photons emitted up to 100 mrad and allow for a free path of photons to the main electromagnetic calorimeter (MEC) emitted up to 7.5 mrad.

HIKE Phase 3 (KLEVER)

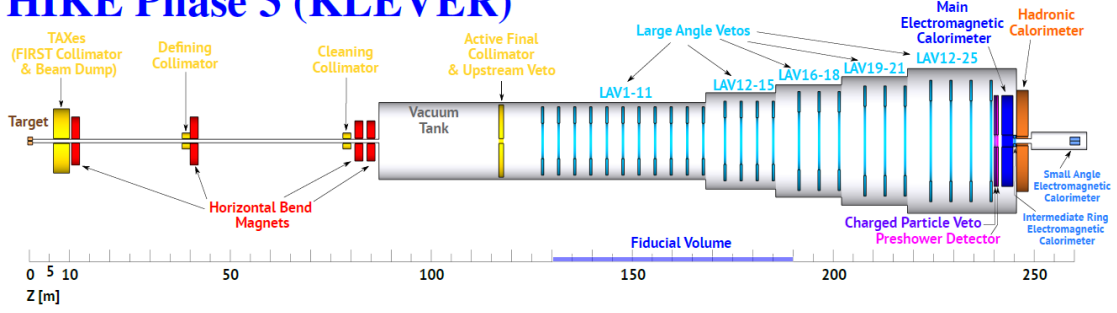


Figure 4: Diagram of HIKE phase 3 detector setup [1]

Due to the differing experimental setup, Phase 3 was separated from the HIKE proposal submitted in 2023 [1]. Still, some detectors would have been shared between the two experiments as the granularity and efficiency requirements for Phase 3 would have driven the detector requirements for the first two phases which would of reduced overall costs. The LAV's, MEC, IRC and SAC and the hadronic calorimeters which would of been built for HIKE Phase 1/2 would of been designed for use in Phase 3. The upstream veto counter, charged particle veto, and the pre-shower detector are unique to Phase 3.

0.2 KTAG Photon Detection System upgrade

To sustain the increased kaon rate, the KTAG required upgrades to the photon-detection system and data acquisition system, with an expected time resolution of 15 - 20 ps and a kaon identification efficiency of greater than 95%. The expected kaon rate was 200 MHz which corresponds to a maximum particle rate $10 \text{ MHz}/\text{cm}^2$

of photons at the PMT array. This was taking into account the production of Cherenkov photons in CEDAR-H and the resultant acceptance of the photons at the PMT plane. It was proposed that the NA62 PMT arrays would be replaced with arrays of micro-channel plate photomultipliers (MCP-PMTs). These PMTs would provide a single photon time resolution of 50 - 70 ps depending on the model, a low dark rate of 1 kHz/cm^2 and the ability to sustain the required photon rate. A downside of the MCP-PMTs is their limited lifetime from the ageing of the Photocathode caused by heavy feedback ions from the residual gas. This results in the rapid degradation of PMT quantum efficiency with the increasing integrated anode charge. MCP-PMTs are similar to normal PMTs except that the dynode section is replaced by a micro-channel plate. An example diagram is shown in Figure ??, where a photon would hit the photocathode window producing a single electron which is then accelerated towards the MCP by an electric field provided by a voltage difference across the gap. The electron then collides with the walls of the MCP producing secondary electrons which results in a cascade which is then collected by the anodes below. The dual layer design or chevron, adds some additional benefits than a single layer one; better gains due to the increased surface for secondary electrons to be produced and improved lifetime due to the suppression of ions produced in the amplification from reaching the photocathode.

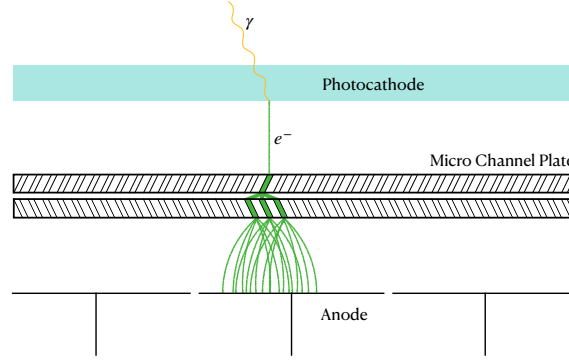


Figure 5: Diagram describing the workings of a MCP-PMT.

An example of the expected MCP-PMT model to be used in the KTAG manufactured by Photonics is shown in Figure 6. The module consists of a 8x8 array of pixels which can individually be readout.

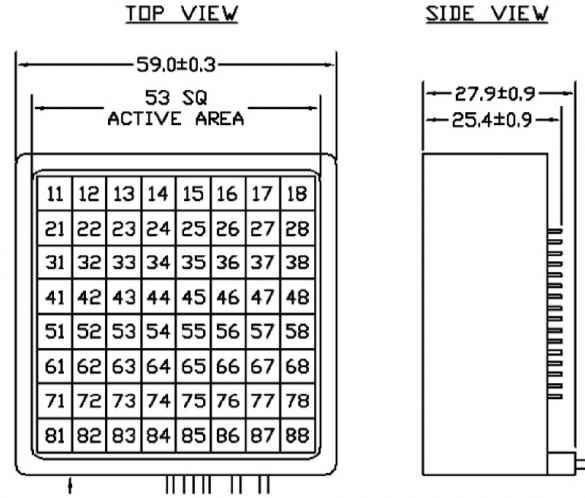


Figure 6: Schematic of Photonics MCP-PMT [5]

Each of the eight sectors would consist of four of these MCP-PMT, shown in Figure 7. The four MCP-PMT per sector were required in order to have full coverage of the light spot produced when the light exiting the CEDAR was reflected by the spherical mirror. As the schematic in Figure 6 shows there is some unused area around the active area of each MCP-PMT, which would result in reduced efficiency,

especially in the centre where the maximum amount of light would be expected. To mitigate this reflective surfaces were attached to each side of the individual MCP-PMTs shown in Figure 7, which reflect the light onto the active surface.



Figure 7: Visualisation of the MCP-PMT array placed inside of the KTAG

The resultant light spot for one of the sectors is shown in Figure 8, where the inactive areas between the individual MCP-PMT are shown and the increased amount of light at the edges from the reflective surfaces. The position on the MCP-PMT in the KTAG sector was optimised to produce the maximum number of photo-electrons per kaon candidate.

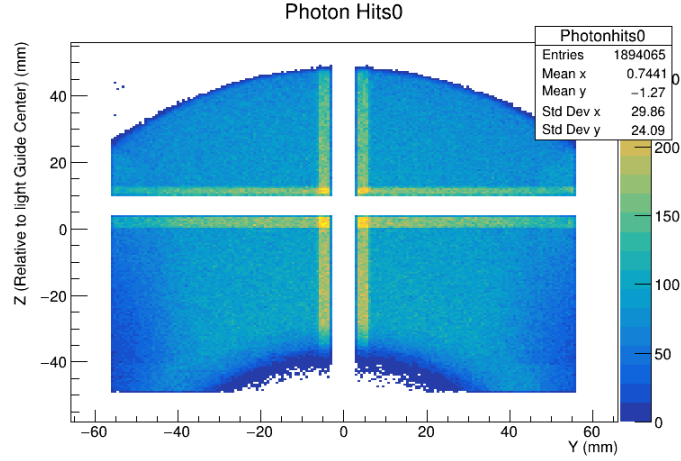


Figure 8: Collected light distribution for the MCP-PMT installed in the KTAG.
(REMAKE)

The Quantum Efficiency of the MCP-PMT from Photonics is shown in Figure 9. There are two options for the photocathode of the MCP-PMT available, a standard bialkali (Sb-Rb-Cs, Sb-K-Cs) or a Hi-QE Blue. The distribution for the Hi-QE Blue was implemented into the Monte Carlo but further studies at The University of Birmingham are planned to take place to characterise the MCP-PMTs by measuring the lifetimes, QE and other characteristics in order to decide which type could be used for high intensity kaon physics at HIKE.

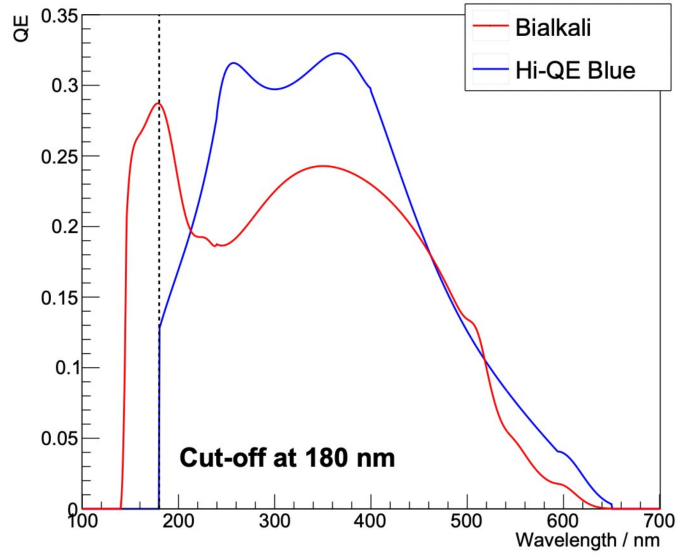


Figure 9: Quantum Efficiency of Photonics MCP-PMT as a function of wavelength (REMAKE, Ask Angela for data)

From the Monte Carlo simulation the implemented MCP-PMTs performed excellently with the result shown in Figure 10. The MCP-PMT produced 24.3 photoelectrons per kaon candidate with a time resolution of 18 ps, taken from the Gaussian sigma of the distribution in Figure 10 (Right). The results surpassed the requirements for the KTAG in HIKE and the upgrades were ready to be used in the HIKE simulations.

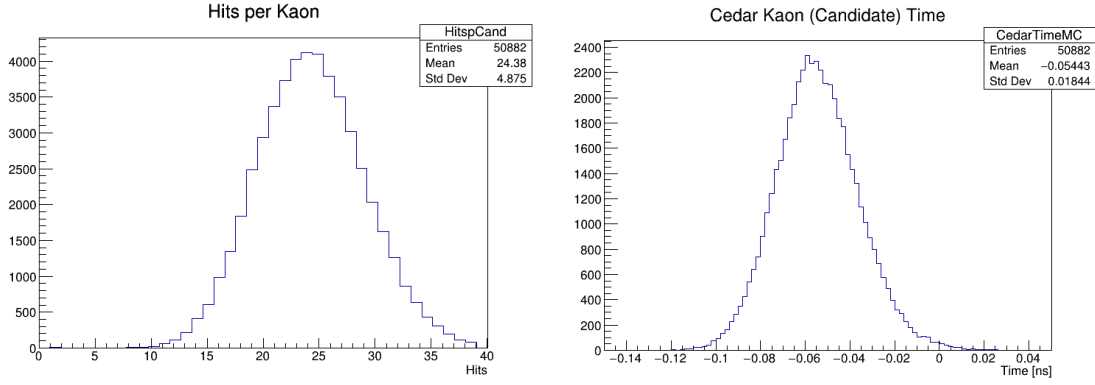


Figure 10: Results from the Monte Carlo simulation of CEDAR-H and MCP-PMT. Left, the number of photo-electrons per kaon candidate and right, the Kaon Candidate time.(REMAKE)

0.3 GTk upgrade design

The NA62 GTk described in Section ?? would have required an upgrade to sustain the fourfold increase in intensity at HIKE, requiring the new detector to have a time resolution of 50 ps compared to 200 ps in NA62. The exact pixel technology was not specified and was not part of this study, with the upgrade being focused on the improved granularity. The NA62 GTk has two different pixel sizes: small, which are $300 \times 300 \times 200 \mu m$, and large, which are $400 \times 150 \times 200 \mu m$. The large pixels are found on the edge of the 10 chips that make up each station, and the small pixels are found in the center shown in Figure 11.

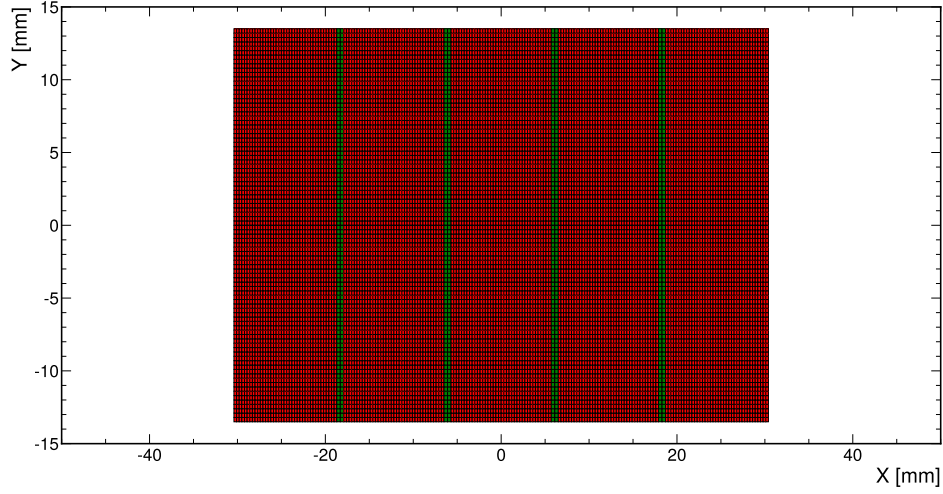


Figure 11: Diagram showing the layout of a NA62 GTK station with small pixels shown in Red and the large pixels shown in green.

Two options were envisioned to fulfil requirements for HIKE, the first having a pixel size on half of NA62 and the second having a pixel size of a third of NA62. Using the upgraded NA62 software package described in Section ?? the two different types of pixels were implemented. For the HIKE simulations used for the proposal, the pixel size of one-third of NA62 was chosen to maximize the granularity of the GTK, with the small pixel size being $150 \times 150 \times 200 \mu m$ and the larger pixel size being $200 \times 150 \times 200 \mu m$. One of the chips of the HIKE GTK station is shown in Figure 12.

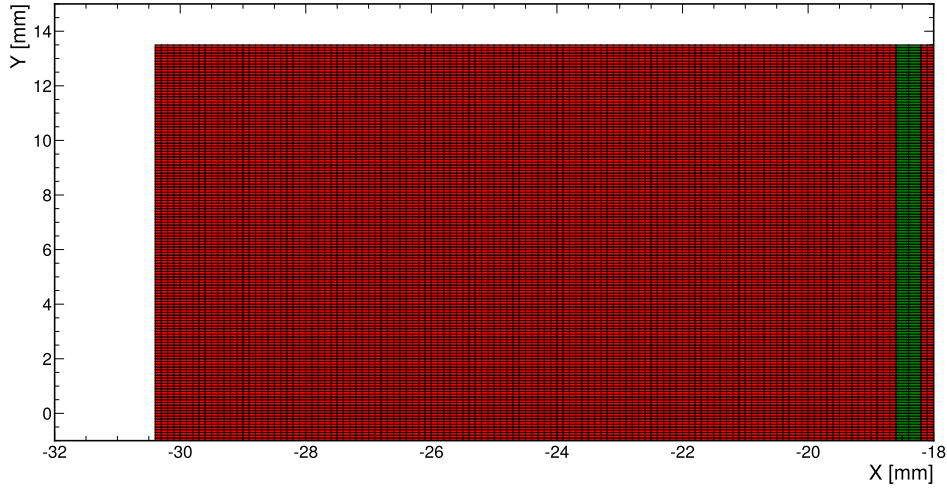


Figure 12: Diagram showing a part of the layout of a HIKE GTK station with small pixels shown in Red and the large pixels shown in green.

0.4 RICH upgrade design

The overall design of the NA62 RICH described in section ?? was well suited for re-use within HIKE with an upgrade to the photon-detection system. The NA62 RICH photon detection system comprises two arrays on either side of the beam pipe, equipped with Hamamatsu R7400-U03 PMTs placed in a pattern described in Figure 13. The PMTs have a single hit time resolution of 240 ps and a quantum efficiency of 20 %. Due to the placement of the PMTs, the distance between the center of each PMT is 18 mm, which is constrained by the sensor size and the geometry of PMT itself. This distance between sensors was the main contributor to the single hit position resolution of 4.7 mm, resulting in an overall ring radius resolution of 1.5 mm. The number of hits per ring candidate and the single hit time resolution produced a time resolution of 80 ps. Due to the increased beam intensity, the main requirement of the RICH for HIKE was a time resolution of 20 - 30 ps.

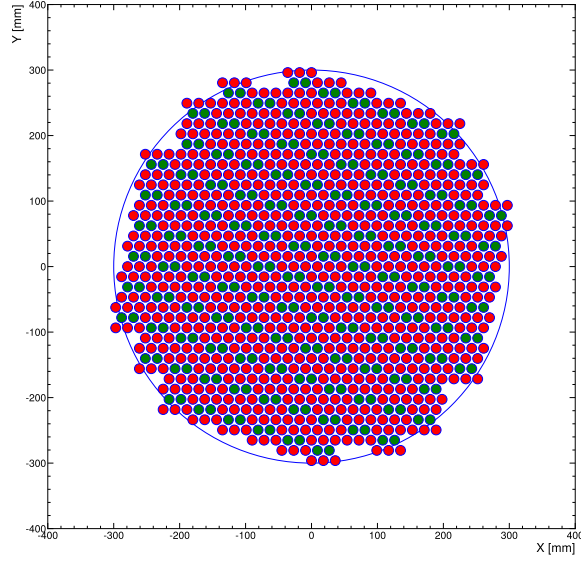


Figure 13: Diagram of RICH PMT array

The proposed RICH photon detection system upgrade would replace the current PMTs with a smaller sensor size, allowing for a smaller distance between sensor centers and reducing the hit position resolution. An excellent candidate for a sensor replacement was silicon photomultipliers (SiPMs), with small sensor sizes, high quantum efficiency of 40% and a single hit time resolution of at least 100 ps were available from major manufacturers. Three sensor sizes were selected for further investigation, $9 \times 9 \text{ mm}$, $6 \times 6 \text{ mm}$, and $3 \times 3 \text{ mm}$. Each size had apparent benefits and downsides, but the smaller sensors would produce a better hit position resolution but at an increased cost and complexity. A study was conducted to determine the optimal sensor size for the SiPMs, with the overall detector performance balanced with the cost and complexity.

Using the upgraded NA62 software package described in Section ??, it was possible to implement the new PMTs into the RICH Monte Carlo simulation with the

implemented arrays shown in Figure 14. Compared to the PMT arrays currently used in NA62, the PMs have been aligned on the vertical axis to simplify the placement because of the square geometry of the new PMTs. The area of the expected light spot, shown in Figure 14 as a grey circle, is completely covered by the SiPMs with no gaps between the sensors which was a clear improvement compared to the NA62 setup shown in Figure 13.

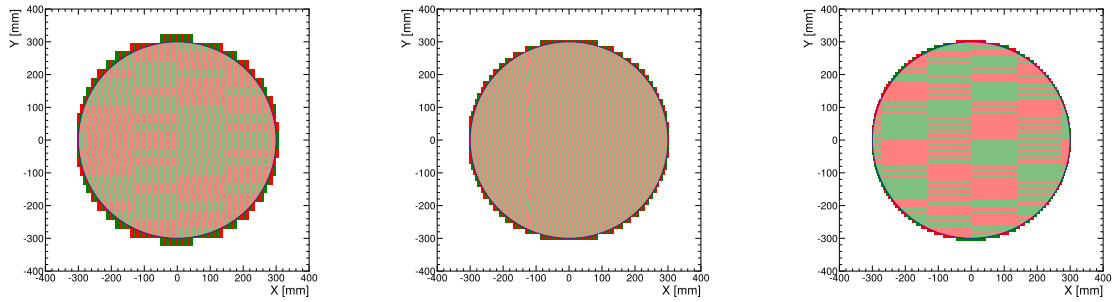


Figure 14: Diagram of proposed RICH SiPM arrays for use in HIKE. Left, 9x9 mm, center 6x6 mm and right 3x3 mm.

The requirements for HIKE proposed a SiPM with a Quantum efficiency of at least 40 % [1]. The Hamamatsu S13360-**50CS series was selected as the preliminary candidate for the RICH PMT upgrade to achieve this requirement. Hamamatsu offers a number of SiPMs in the S13360 series, with the **50CS selected for its improved Ultra Violet (UV) sensitivity and high quantum efficiency of 40 % at 460 nm, shown in Figure 15.

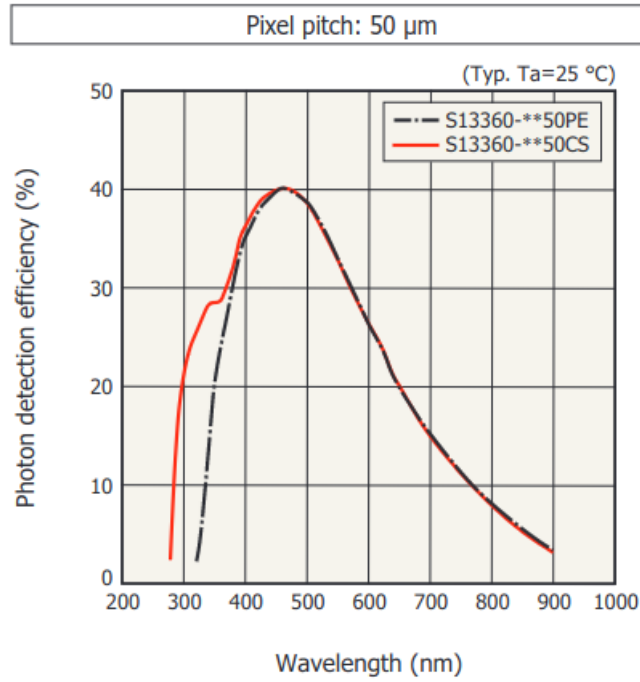


Figure 15: Selected SiPM quantum efficiency taken from Hamamatsu S13360 data-sheet [6]

Along with the QE efficiency, a single hit time resolution of 100 ps was required [1]. The manufacturer stated the Hamamatsu S13360-**50CS series has excellent time resolution, but no definite value was given. The time resolution would also depend on the final design of the RICH readout electronics and would have to be measured in a test setup. A simple Gaussian time response curve with a sigma of 100 ps was used for the simulations as a placeholder, ready for a specific model to be implemented when chosen.

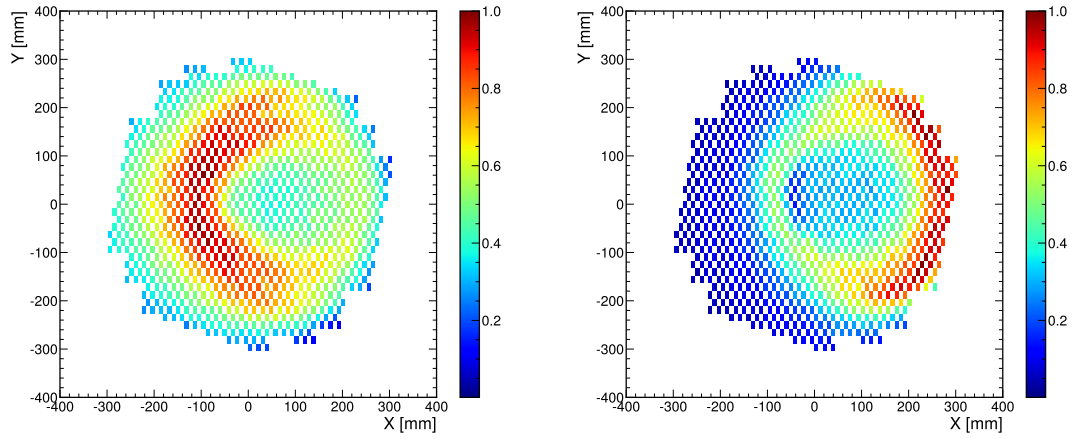
The expected performance of the RICH detector with the new PMTs was compared to the current NA62 RICH detector in Table 1. With the increased quantum efficiency, the expected number hits per π^+ between 15 and 45 GeV/c was expected to increase by a factor of two. This, along with the improved time resolution of the new PMTs resulted in an expected time π^+ time resolution of 27 ps at 15 GeV/c

	NA62 RICH	HIKE RICH
Sensor time resolution	240ps	100ps
Sensor quantum efficiency	20%	40%
Number of photo-electrons for π^+ at 15 GeV/c	7	14
Number of photo-electrons for π^+ at 45 GeV/c	12	24
Time resolution for π^+ at 15 GeV/c	90ps	27ps
Time resolution for π^+ at 45 GeV/c	70ps	20ps

Table 1: Comparisons of the expected performance of the NA62 and HIKE RICH detectors [1]

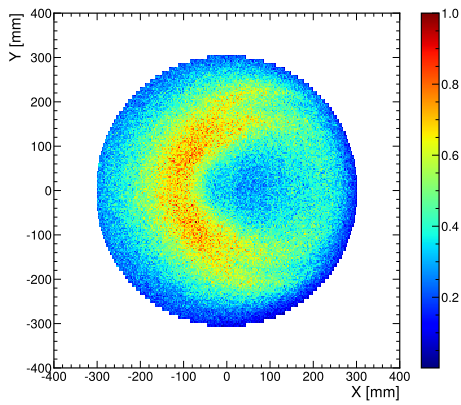
and 20 ps at 45 GeV/c . Using the implemented upgrades to the RICH detectors, the above expectations were confirmed by a Monte Carlo sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3).

For the three sensor sizes, a test production of 1M Ke3 was used to determine the preliminary performance of the detector and compare this to the current NA62 setup. The plots in Figure 17 show the light distribution on the PMT arrays for the three sensor sizes: 3 mm 17(a,b), 6 mm 17(c,d), and 9 mm 17(e,f) compared to the illumination on the current NA62 PMT arrays 16. Due to the better placement of the square sensors, it is possible to see a more uniform illumination of the PMT arrays and better coverage of the light ring, especially in the Jura side. It is clear to see that either sensor size would be an improvement on the current NA62 setup.

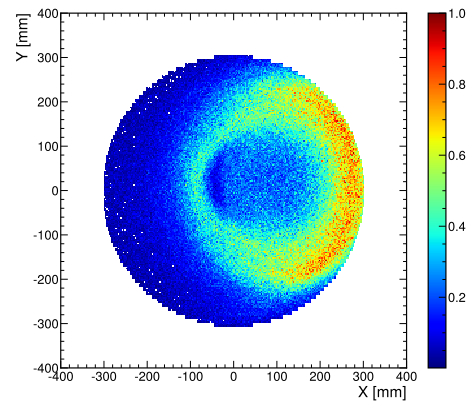


(a) Light distribution for NA62 setup on the Saleve side of the RICH detector. (b) Light distribution for NA62 setup on the Jura side of the RICH detector.

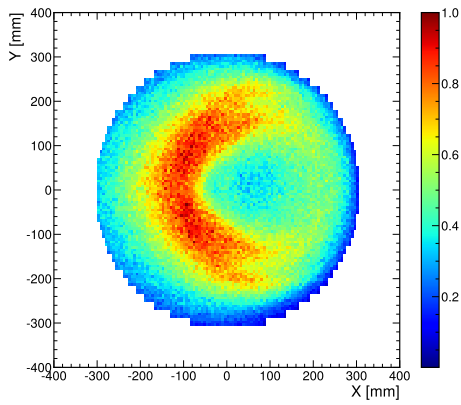
Figure 16: Cherenkov light illumination of RICH PMT arrays for the NA62 PMT arrays produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3). The left column shows illumination on the Saleve side of the RICH detector, and the right column shows the Jura side.



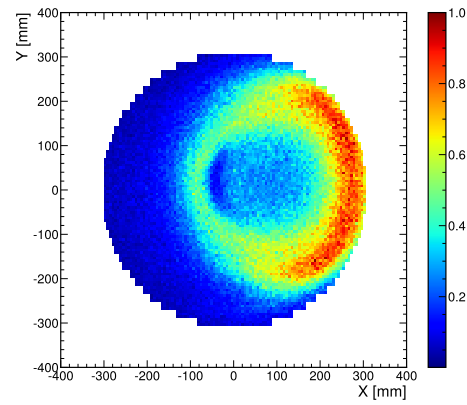
(a) 3 mm Saleve.



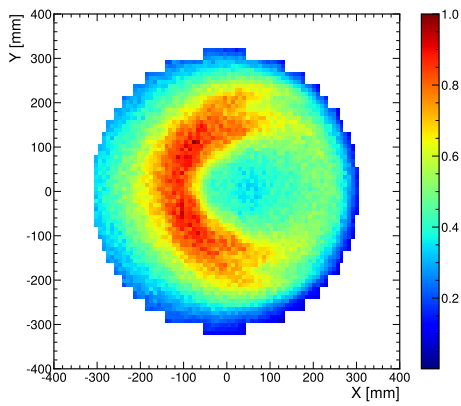
(b) 3 mm Jura.



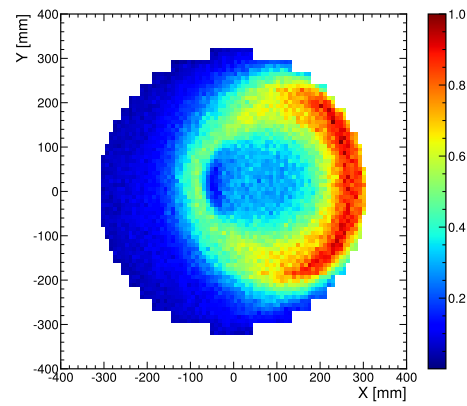
(c) 6 mm Saleve.



(d) 6 mm Jura.



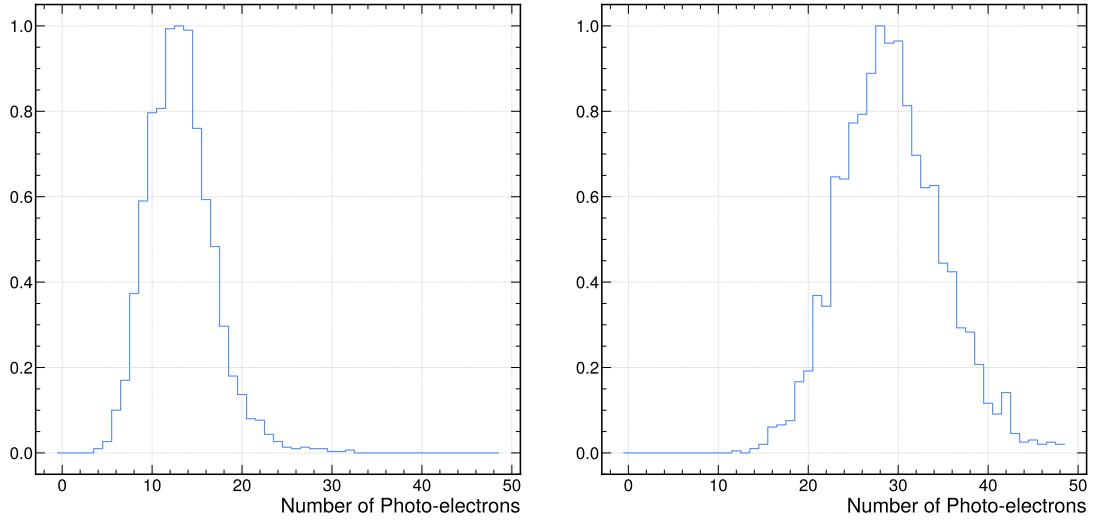
(e) 9 mm Saleve.



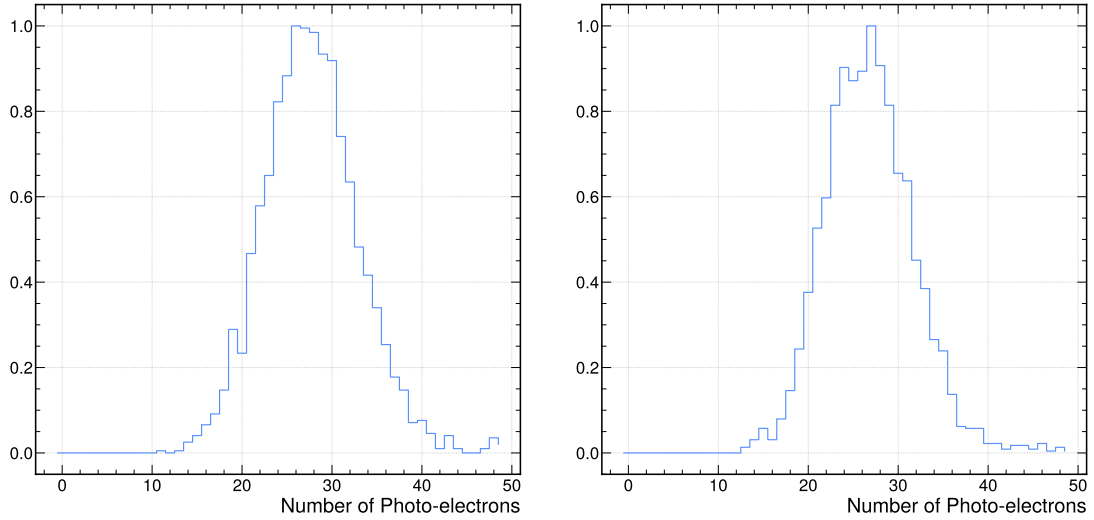
(f) 9 mm Jura.

Figure 17: Cherenkov light illumination of RICH PMT arrays for the three different sensor produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3). The left column shows illumination on the Saleve side of the RICH detector, and the right column shows the Jura side.

In the reconstruction the photo-electrons are clustered into ring candidates. The number of reconstructed photo-electrons is shown in Figure 18, with the SiPM sensors resulting in a two-times increase in the number of hits per ring candidate. The improved number of reconstructed photo-electrons was mainly due to the enhanced quantum efficiency of the SiPMs and the sensors' better placement, reducing the probability of multiple photons hitting the same sensor. There are minimal differences between the three sensor sizes shown in Figure 18 (b-d), with the 3 mm sensors having a slightly higher number of hits per ring candidate when compared to the 9 mm sensors.



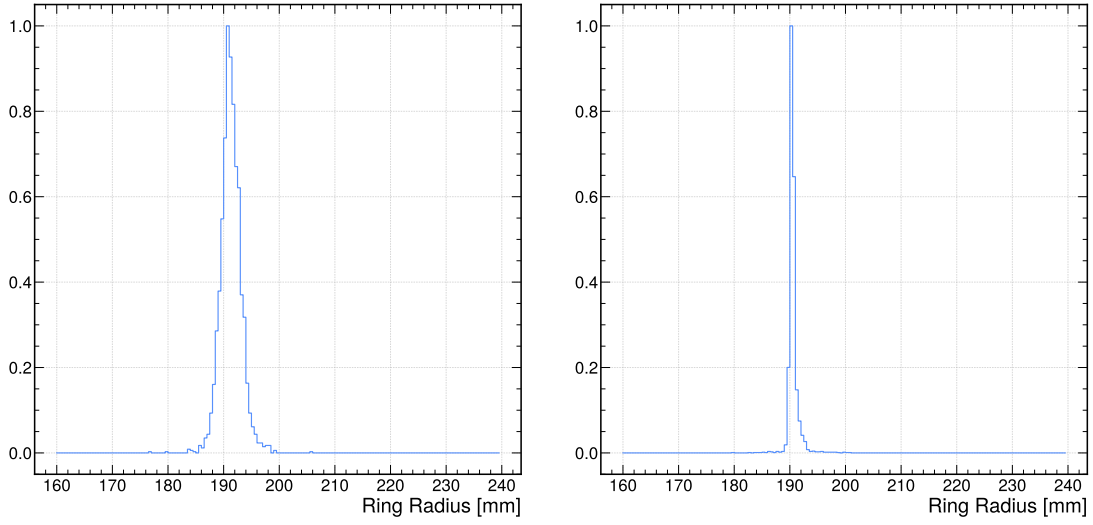
(a) Number of hits per ring candidate for NA62 PMT arrays, $\mu = 13.00 \pm 0.01$ (b) Number of hits per ring candidate for 3 mm SiPMs, $\mu = 28.86 \pm 0.01$.



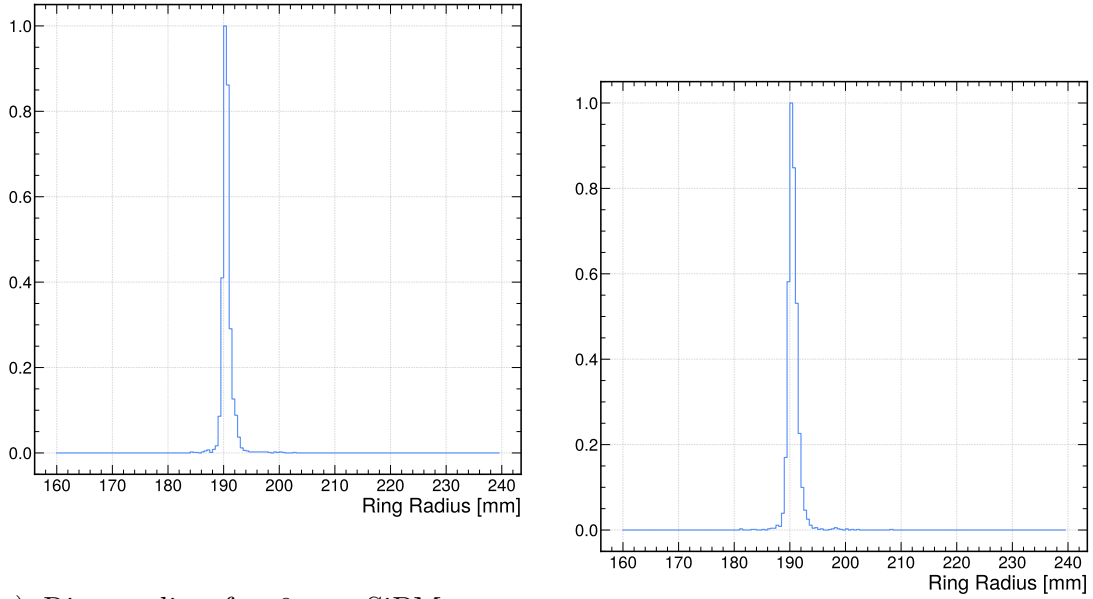
(a) Number of hits per ring candidate for 6 mm SiPMs, $\mu = 27.56 \pm 0.01$. (b) Number of hits per ring candidate for 9 mm SiPMs, $\mu = 26.36 \pm 0.01$.

Figure 18: Number of photo-electrons per ring candidate for NA62 PMTs and the three possible HIKE SiPMs produced with a from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3).

The subsequent reconstructed ring candidate radius is shown in Figure 19, with the NA62 PMT array producing a resolution of 1.68 mm compared to 0.72 mm for the HIKE array with 9 mm SiPMs. This was a over two times increase in resolution with the largest sensor size with the 3 mm improving further to 0.41 mm. However it was noted that this resolution would not be possible in data-taking due to inaccuracies with the mirror placement and dispersion in the radiator gas. It was accepted that the 9 mm value would be the best possible resolution obtainable.



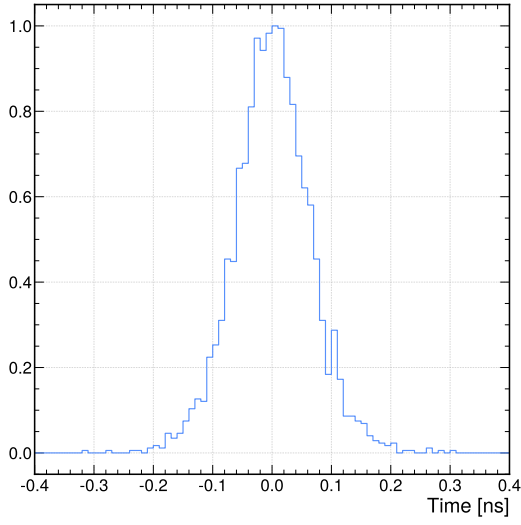
(a) Ring radius for NA62 PMT arrays, (b) Ring radius for 3mm SiPMs, $\sigma = 1.68 \pm 0.03$. 0.41 ± 0.01 .



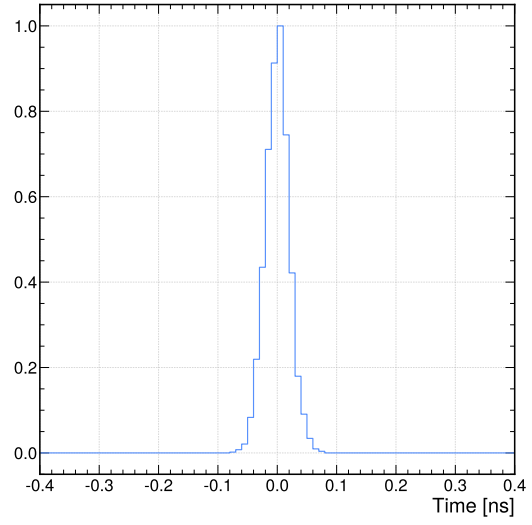
(a) Ring radius for 6mm SiPMs, $\sigma = 0.56 \pm 0.01$. (b) ν for 9mm SiPMs, $\sigma = 0.72 \pm 0.01$.

Figure 19: Reconstructed ring radius for electron candidates for NA62 PMTs and the three possible HIKE SiPMs produced with a from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (Ke3).

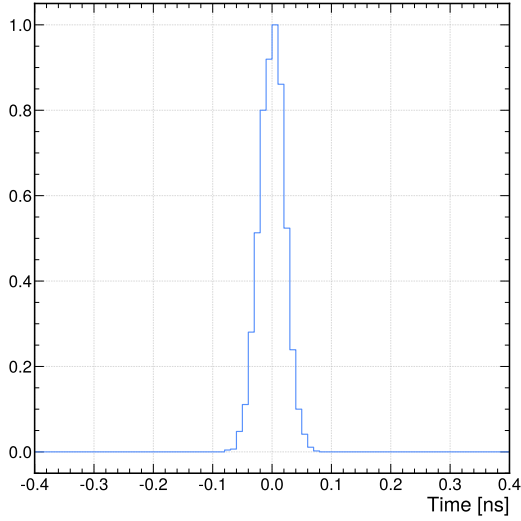
The time resolution of the RICH detector was calculated using the Ke3 sample by equally splitting the reconstructed photo-electrons of the ring candidates into two sets and calculating the time difference between them, with the results are shown in Figure 20. The 3x3mm sensors produced the best time resolution of 20.04ps compared to the 9x9mm sensors with a time resolution of 21.22ps. However there was little difference between the SiPMs with each sensor size producing a time resolution of less than 30 ps, fulfilling the requirements of HIKE. The improvement was clear with using the small SiPMs when compared with the NA62 resolution in Figure 20 which had a time resolution of 61.35 ps.



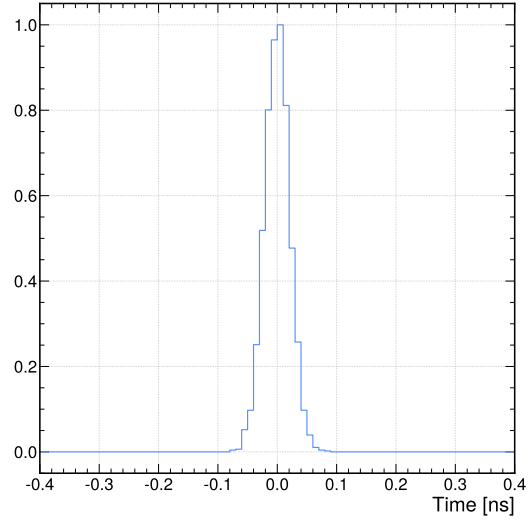
(a) Time resolution for NA62 RICH.



(b) Time resolution for 3 mm SiPMs.



(a) Time resolution for 6 mm SiPMs.



(b) Time resolution for 9 mm SiPMs.

Figure 20: Time resolution for NA62 PMTs and the three possible HIKE SiPMs, calculated from the time difference between two equally populated subset of hits within the ring candidate. A detailed description of the procedure is provided here [7]

After the RICH MC study it was proven that the introduction of small SiPMs

would dramatically increase the granularity of the detector. All three SiPM sensor sizes produced results which exceeded the requirements for HIKE with time resolution of around 20 ps and improvements on the reconstructed ring radius resolution. It was decided that the 9x9 mm SiPMs would be used for the HIKE simulations used for the proposal as they provided excellent performance and would reduce the cost and complexity with implementing the smaller sensor sizes in the detector when it was built. A clear benefit was the reduction of number of channels from 72780 for the 3x3 mm compared to 8588 for the 9x9 mm SiPMs.

0.5 Detector Rates

After the new GTK and RICH detectors were developed it was necessary to determine the expected particle rate in the upgraded detectors and the other detectors. For the expected rate for Phase 1, it was possible to estimate by scaling the known NA62 rates by the expected increase in beam intensity, by a factor of 4 compared to NA62. The rates for Phase 2 were calculated using the new simulation software described in the previous sections.

An array of input files were produced in BDSIM [8] by the CERN beam group which contained the simulated HIKE beam components at a Z of 120 m which was the start of the fiducial volume. The files were then used as an input for the NA62MC software and used as the primary beam instead of the turtle beam which is default for NA62. Each file contained 10000 events which was equivalent to 10000 protons on target and there were 18K files produced in total which was equivalent to 1.8×10^8 protons on target. As this wouldn't be enough to get a good estimate of the rates, each file was used 11 times as an input to the simulation with random seeds to

increase the statistics. The simulation was ran with the unforced flag which allows for four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$. After simulation the files were then reconstructed using the NA62 Reconstruction software and the rates were calculated by scaling the numbers in the output by the expected HIKE Phase 2 beam conditions, a spill time of 4.8 seconds and a beam intensity of 2×10^{13} protons per spill.

The K_L rate at the entrance of the fiducial volume was calculated to be 304.1 MHz (ADD ERROR), with the rate of the four decays in the fiducial volume shown in Table 2.

KL decay	Rate [MHz]
$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0)	13.89
$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$)	9.63
$K_L \rightarrow \pi^0 \pi^0 \pi^0$	6.60
$K_L \rightarrow \pi^+ \pi^- \pi^0$	4.45

Table 2: Rates of the four main K_L decays in the HIKE phase 2 fiducial volume. (ADD ERROR)

The total rates from all particles in the events where the K_L decays are shown in Table 3. To calculate the rate all hits in all of the channels of each detector were collected and converted to the expected rate. The rates were large but this was to be expected with a high intensity, however the rate in the spectrometer for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay was concerning as one would expect a much lower rate compared to the other decays due to the lack of a charged daughter particles. Even with a lower branching ratio, it is expected that the $K_L \rightarrow \pi^+ \pi^- \pi^0$ decay should have a

higher rate in the spectrometer due to the charged pions.

Detector	K_L Decay [MHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$	$K_L \rightarrow \pi^0 \pi^0 \pi^0$	$K_L \rightarrow \pi^+ \pi^- \pi^0$
	(40.55 $\pm$ 0.11)%	(27.04 $\pm$ 0.07)%	(19.52 $\pm$ 0.12)%	(12.54 $\pm$ 0.05)%
STRAW-1	126.8	79.9	50.5	49.2
STRAW-2	162.8	85.1	80.7	58.0
STRAW-3	165.7	89.3	67.7	56.1
STRAW-4	189.2	93.6	87.5	64.4
NCHOD	10.3	5.2	4.9	3.7
CHOD	90.5	33.4	61.3	33.8
IRC	3.8	1.2	4.2	2.0
LKr	609.5	266.0	440.7	298.3
MUV1	129.9	102.4	0.6	64.1
MUV2	10.3	17.9	0.1	4.7
MUV3	0.9	5.2	0.2	0.6
HAC	10.0	11.5	0.3	6.1

Table 3: Total rates in the detectors for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$.

Directly looking at the MC it was possible to find the origin of the particles which produced the hits in the spectrometer. The production Z compared against the production radius of particles which produced a hit in the Spectrometer is shown in Figure 21. It is clear to see the for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay, the majority (80 %) of the particles were produced after Z=210 m with most produced in the walls of the vacuum decay tube and in the spectrometer chambers themselves, with only 20 % of

particles produced before $Z=210$ m which would be consistent with being produced in the decay.

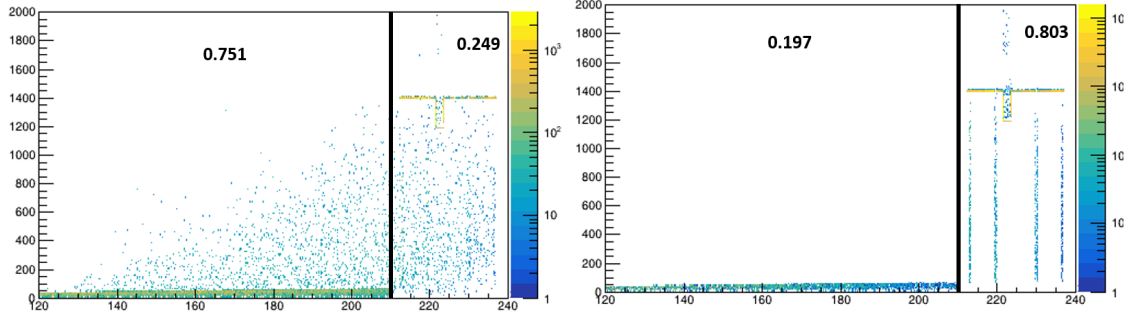


Figure 21: Production Z compared against production radius of particles producing a hit in the Spectrometer with the fraction of particles produced before and after $Z=210$ m. Left, for the $K_L \rightarrow \pi^+ \pi^- \pi^0$ decay and right, for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay.

Again this is clear in Figure 22 where the MC particle identification code of the particles producing hits in the spectrometer is shown. For the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ the majority of the particles are electron/positron pairs, produced in interactions of photons in the walls of the vacuum decay tube and in the spectrometer chambers, compared to the $K_L \rightarrow \pi^+ \pi^- \pi^0$ decay where the majority of the particles are pions produced in the decay.

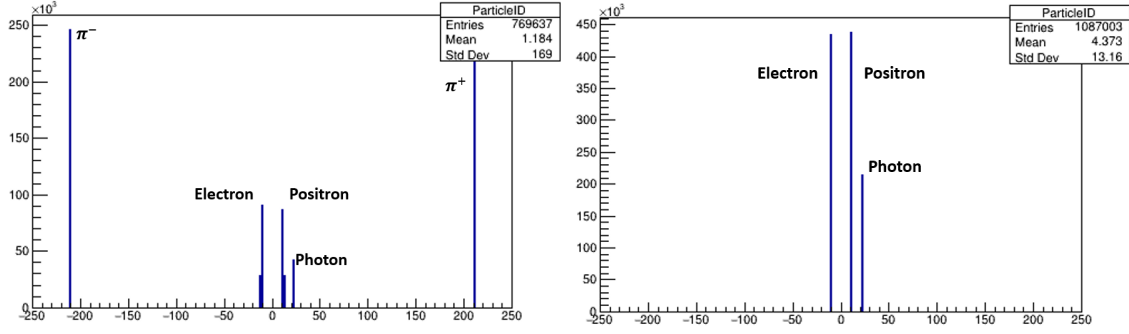


Figure 22: Monte Carlo particle code for the particles producing a hit in the Spectrometer. Left, for the $K_L \rightarrow \pi^+ \pi^- \pi^0$ decay and right, for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay.

As well as the overall hit rate, it is important to understand the rate in the detectors produced directly from the daughter particles of the K_L decays. The daughter particles were selected as particles with a GEANT4 production process of "Decay", which should include any subsequent decays of the daughter particles eg $\pi^0 \rightarrow \gamma\gamma$. This should exclude all hits produced from particles produced by "gamma conversion" which produce electron positron pairs and impact the rate, as shown above. The rates for each detector is shown in Table 4. It was clear that the overall rates were much lower when compared to the total rates in Table 3 and that the rates in the spectrometer for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay are much lower which is expected. Overall the rates were acceptable for the detectors as it was assumed that most of the hits produced not from the decay of the K_L or other subsequent decays would be removed by the trigger system as they would be outside the time window of the K_L decay.

Detector	K_L Decay Particle Rates [MHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	14.7	11.2	1.3	5.6
STRAW-2	15.4	11.7	2.1	6.2
STRAW-3	16.1	12.1	2.5	6.6
STRAW-4	16.5	12.2	3.4	6.9
NCHOD	19.9	10.8	9.4	7.0
CHOD	16.1	11.3	8.1	7.8
IRC	1.9	0.8	1.8	1.0
LKr	15.4	11.6	19.9	10.5
MUV1	7.6	9.8	0.3	4.7
MUV2	2.7	7.2	0.1	1.5
MUV3	1.1	6.7	0.3	0.9
HAC	0.6	1.0	0.1	0.3

Table 4: Detector rates for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

Rates for the detectors were important to understand in order to determine the overall detector design, especially for the electronics that would need to be designed to meet the high rates. Along with the overall rate it was vital to know the max channel rates in the detectors to design the electronics to handle the highest rate expected. The max channel rates for each detector were collected and are shown in Table 5.

Detector	K_L Decay Daughter Max Channel Rates [kHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$	$K_L \rightarrow \pi^0 \pi^0 \pi^0$	$K_L \rightarrow \pi^+ \pi^- \pi^0$
	$(40.55 \pm 0.11)\%$	$(27.04 \pm 0.07)\%$	$(19.52 \pm 0.12)\%$	$(12.54 \pm 0.05)\%$
STRAW-1	325	254	51	186
STRAW-2	377	254	27	154
STRAW-3	329	265	19	138
STRAW-4	365	258	39	138
NCHOD	484	329	186	206
CHOD	1290	635	786	559
IRC	1020	345	1063	516
LKr	575	321	385	293
MUV1	1830	1496	19	1032
MUV2	329	583	15	190
MUV3	79	234	39	43
HAC	206	210	27	138

Table 5: Max channel rates for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

As detailed previously in this section the upgraded detectors in the simulation were the RICH the GTK along with the upgraded STRAW spectrometer. This allowed for correct rate calculation for all three of these detectors, however it was foreseen that all detectors would be upgraded to increase the granularity and the rate handling capabilities. This would result in an incorrect calculation as the rate is determined by the NA62 detector design and the layout of the channels

in the detectors. As the final designs for these detector were not finalised, it was impractical to implement the new detectors in the simulation. However, it was still possible to calculate the rates at the planes of the upgraded detectors by collecting particles passing through the Z of the detector. This was possible using raw MC where information for particles are stored at different parts of the HIKE setup (checkpoints). The rates for charged particles produced from the four K_L decays at each checkpoint are shown in Table 6.

Checkpoints (Z Position)	K_L Decay Particle Rates [kHz/cm ²]							
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$		$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$		$K_L \rightarrow \pi^0 \pi^0 \pi^0$		$K_L \rightarrow \pi^+ \pi^- \pi^0$	
	(40.55 ± 0.11)%		(27.04 ± 0.07)%		(19.52 ± 0.12)%		(12.54 ± 0.05)%	
	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	19.28	0.34	16.64	0.25	0.05	10 ⁻³	10.55	0.13
STRAW-3 Entry (230.5 m)	6.93	0.29	6.23	0.23	0.14	0.02	4.04	0.12
NCHOD Entry (237.6 m)	4.93	0.16	4.43	0.13	0.17	0.01	2.66	0.06
CHOD Entry (238.9 m)	4.60	0.27	4.07	0.21	0.14	0.02	2.55	0.11
IRC Entry (239.5 m)	4.77	0.26	3.89	0.21	0.13	0.02	2.43	0.11
LKr Entry (240.4 m)	4.60	0.15	3.72	0.13	0.10	0.01	2.34	0.06
MUV1 Entry (243.2 m)	3.92	0.02	3.38	0.08	0.03	10 ⁻⁴	1.96	0.02
MUV2 Entry (244.4 m)	3.70	0.01	3.25	0.08	0.03	10 ⁻⁴	1.82	0.01
MUV3 Entry (245.3 m)	0.15	0.08	1.05	0.08	0.01	10 ⁻⁷	0.09	0.01
HAC Entry (253.3 m)	3.40	0.09	2.88	0.09	0.02	10 ⁻⁵	1.44	0.01

Table 6: Particle rates at specified checkpoints along the HIKE detector setup. Including the average rate across the cross section and the maximum rate per cm² for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

However as the rates are collected from the cross section at the expected detector

position, the rates are higher than what would be seen due to the fact that all the detectors would eventually have a beam hole to accommodate beam particles passing through the detector. An example cross section at the entry of first STARW chamber is shown in Figure 23, where it is clear that the highest rate from the decay particles was found at the centre. As the detectors were not designed and the beam hole size was not known, it was possible to estimate the estimated size by calculating the beam divergence at the detector position. This was calculated using Equation 0.1.

$$\delta_{beam} = 0.4 \text{ mrad} \times Z \quad (0.1)$$

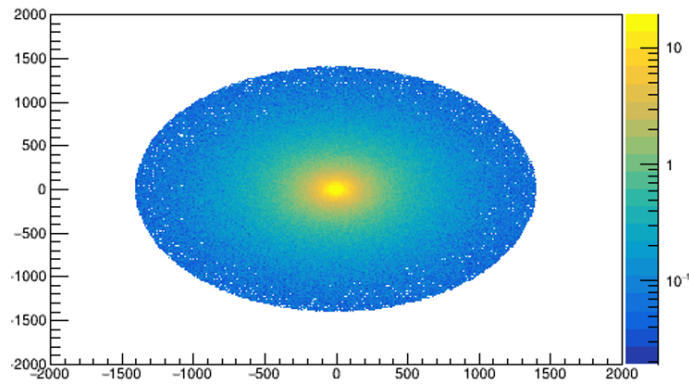


Figure 23: Rate of particles across the cross section at the first STRAW chamber entrance at $Z=213.\text{m}$.

Using Equation 0.1, a mask for where the beam hole would be was applied to remove any particles that would not impact the detector. The resultant rates with the beam hole mask applied are shown in Table 7. As expected this reduced the max rate for most detectors without impacting the average rate across the detector plane.

0.5. DETECTOR RATES

		K_L Decay Particle Rates [kHz/cm ²]							
		$K_L \rightarrow \pi^\pm e^\mp \nu_e$		$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$		$K_L \rightarrow \pi^0 \pi^0 \pi^0$		$K_L \rightarrow \pi^+ \pi^- \pi^0$	
		(40.55 ± 0.11)%		(27.04 ± 0.07)%		(19.52 ± 0.12)%		(12.54 ± 0.05)%	
Checkpoints (Z Position)	Beam radius [mm]	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	85.4	11.59	0.305	9.504	0.226	0.047	0.001	5.863	0.109
STRAW-3 Entry (230.5 m)	92.2	5.515	0.280	4.900	0.219	0.139	0.017	3.087	0.113
NCHOD Entry (237.6 m)	95.0	3.959	0.152	3.143	0.124	0.166	0.012	2.032	0.063
CHOD Entry (238.9 m)	95.6	3.418	0.256	3.017	0.202	0.139	0.017	1.849	0.104
IRC Entry (239.5 m)	95.8	3.418	0.249	2.921	0.198	0.133	0.016	1.733	0.100
LKr Entry (240.4 m)	96.2	2.501	0.145	2.053	0.122	0.100	0.010	1.314	0.059
MUV1 Entry (243.2 m)	97.3	0.279	0.016	1.311	0.074	0.033	10 ⁻⁴	0.227	0.012
MUV2 Entry (244.4 m)	97.8	0.167	0.008	1.215	0.069	0.027	10 ⁻⁴	0.116	0.007
MUV3 Entry (245.3 m)	98.1	0.153	0.008	1.051	0.079	0.007	10 ⁻⁷	0.089	0.007
HAC Entry (253.3 m)	101.3	0.167	0.008	1.147	0.078	0.013	10 ⁻⁶	0.138	0.007

Table 7: Particle rates at specified checkpoints along the HIKE detector setup excluding particles passing within the expected beam radius for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

For the detector design it was also important to calculate the total rates produced by different particles, which is shown in Table 8. Its clear that the majority of the rates in the at the detector planes was from photons, especially in the spectrometer. However, the rate from charged particles which the detectors would be mainly designed for is reasonable suggesting that the detectors could be designed and built with current technologies.

With all rates information from the simulation shown in this section, it would then possible to design the detectors that were not already upgraded within the simulation. The Table 8 was used in the HIKE proposal to demonstrate the requirement for upgrading the detectors to handle the increased rates.

Checkpoints	Total Particle Rates kHz/cm ²									
	$e^{+/-}$		$\mu^{+/-}$		$\pi^{+/-}$		γ		Charged Particles	
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	264	0.26	224	1.58	117	0.41	13426	13.87	360	2.56
STRAW-3 Entry (230.5 m)	246	0.59	179	1.34	120	0.40	11076	14.02	291	2.62
NCHOD Entry (237.6 m)	216	0.45	174	1.13	115	0.24	11770	8.50	238	1.98
CHOD Entry (238.9 m)	318	2.96	176	1.18	112	0.41	11687	15.57	376	5.79
IRC Entry (239.5 m)	251	3.85	176	1.17	85	0.40	11481	15.58	291	5.64
MEC Entry (240.4 m)	85	1.76	174	1.12	109	0.24	166	0.30	238	3.25
MUV1 Entry (243.2 m)	5	0.01	190	1.04	32	0.03	29	0.01	190	1.07
MUV2 Entry (244.4 m)	8	10^{-3}	162	1.00	5	10^{-3}	5	10^{-3}	163	0.98
MUV3 Entry (245.3 m)	5	10^{-4}	115	0.89	3	10^{-4}	3	10^{-3}	115	0.88
HAC Entry (253.3 m)	11	0.02	120	0.89	5	10^{-3}	3	10^{-3}	120	0.89

Table 8: Average and Max particle rates per cm^2 for specified checkpoints along the length of the detector. The particle rates are collected not from the geometrical acceptance of the corresponding sub-detectors but from the cross section of the outer volume for the sub-detector found in the Monte Carlo. For the Straw-1&3 a circle of radius of 141.5 mm, NCHOD a square of length 330 mm, CHOD a square of length 244 mm, IRC a square of length 122 mm, MEC a circle of radius 182.5 mm, MUV-1&2 a square of length 320 mm, MUV-3 a square of length 281 mm and HAC a square of length 281 mm. A geometrical cut has been made to remove particles that pass through the checkpoints within the expected beam dispersion radius calculated to be $0.4 \times 10^{-3} \times Z$, where Z is the position of the checkpoint.

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